

QIMING CUI

Applied Mathematics and Statistics Dept., Johns Hopkins University, Maryland, USA 21218

☎ (+1) 667-263-9069 ✉ qcui6@jhu.edu  [LinkedIn](#)

CANDIDATE INFORMATION

- 5th year Ph.D. Candidate in Applied Mathematics and Statistics Department, Johns Hopkins University
- Research area: Algorithms with machine-learned predictions (learning-augmented algorithms)
- Expertise: Algorithmic decision-making under uncertainty, online algorithms, and optimization.

EDUCATIONAL BACKGROUND

Johns Hopkins University

Doctor of Philosophy in Applied Math and Statistics

August 2027 (expected)

Adviser: Dr. Michael Dinitz

University of Oxford

Master of Science in Mathematical Sciences

July 2022

Graduate with Merit

University of Liverpool

Bachelor of Science with Honours in Mathematics (First Class)

June 2021

GPA: 4.0/4.0

RELEVANT COURSEWORK

- | | | | |
|--------------------------|---------------------------|------------------------------|----------------------|
| • Algorithm Design | • Machine Learning Theory | • Convex Optimization | • Data Science |
| • LLM foundations | • Probability Theory | • Continuous Optimization | • Matrix Analysis |
| • Reinforcement Learning | • Statistics Theory | • Combinatorial Optimization | • Numerical analysis |

RESEARCH PROJECT EXPERIENCES

Contention resolution with distributional predictions

- Paper submitted to SIAM Symposium on Simplicity in Algorithms (SOSA 2027)
- Designed new algorithms, proved technical upper bounds and gave essentially matching lower bounds

Ski rental with distributional predictions of unknown quality

- Paper accepted at International Conference on Machine Learning (ICML 2026)
- Designed robust algorithms to the classical ski rental problem with any distributional prediction
- Proved high-level upper bound and graceful degradation of algorithm performance with unknown increasing error
- Developed consistency-robustness guarantees, Pareto-optimality results and tight lower bounds.

Controlling tail risk in two-slope ski rental

- Paper accepted at Approximation and Online Algorithms (WAOA 2025) and invited to special issue of Acta Informatica
- Developed technical theorems to fully understand the structure of optimal solution(s) for the problem
- Designed two algorithms to compute the optimal solution(s) for the problem

β -expansions

- Finished a final year report about theorems and properties of β -expansions from dynamical systems viewpoint
- Exploited the number of G -expansions, established a sophisticated theorem and gave pages of formal proofs
- Studied Uniqueness expansions of points, provided new proofs of lemmas and theorems

PROFESSIONAL SKILLS

- Programming Languages: Python (Pandas, Numpy)
- Office Tools: Microsoft Office (Excel, Word, PowerPoint), Google Workspace (Docs, Sheets, Slides)
- Math Tools: LaTeX, MATLAB

HONORS AND AWARDS

- Bronze medal in 28th International Mathematics Competition for University Students
- 2019-2020, 2020-2021 University of Liverpool Excellence Scholarship (1/320)
- 2019 Summer Undergraduate Research Fellowship

EXTRACURRICULAR / ACADEMIC SERVICE

- Co-instructor of the course EN.553.803 Financial Computing Workshop (Intro to Agentic AI) at JHU
- Research Assistant in Algorithms and Complexity group at the Johns Hopkins University
- Teaching Assistant in Machine Learning (EN.553.740), Computing for Applied Math (EN.553.688), Approximation Algorithms (EN.601.635) and Numerical Linear Algebra (EN.553.385)
- University of Oxford Mathematics Admission Test (MAT) Grader

PERSONAL INTERESTS

- World travel, natural landscapes, intellectual exploration, and thoughtful reflection